



HABIB UNIVERSITY
Digital Logic and Design
EE-172 / CS-130 / CE-222
Fall 2025 (L2)

Quiz # 4

Student ID: _____

Student Name: _____

Date: November 20, 2025

Time: 30 minutes

Max points: 16

Instructions:

- Smart watches, laptops, and similar electronics are strictly NOT allowed.**
- Answer sheets should contain all steps, working, explanations, and assumptions.**
- Attempt the quiz in the spaces provided for each question.
- You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

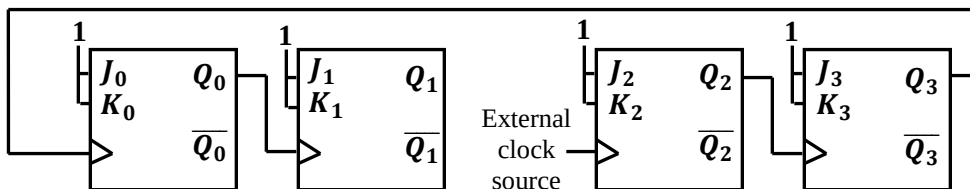
No.	Questions	Pts
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- | | | |
|---|--|---|
| 1 | Consider a positive edge triggered JK flip flop. Preset and clear inputs are active high. Indicate the output for each of the following cases. | 4 |
|---|--|---|

Clock	Preset	Clear	Current state	J	K	Next state
↑	0	0	1	0	0	1 (no change)
1	0	0	0	0	1	0 (no change)
↓	1	0	0	1	0	1 (preset)
↑	0	1	1	1	1	0 (clear)

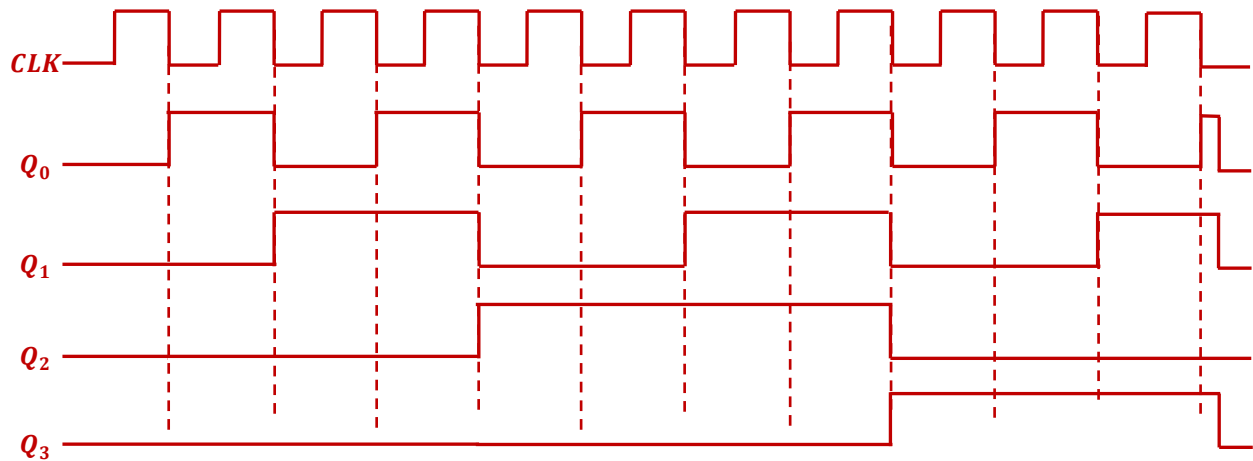
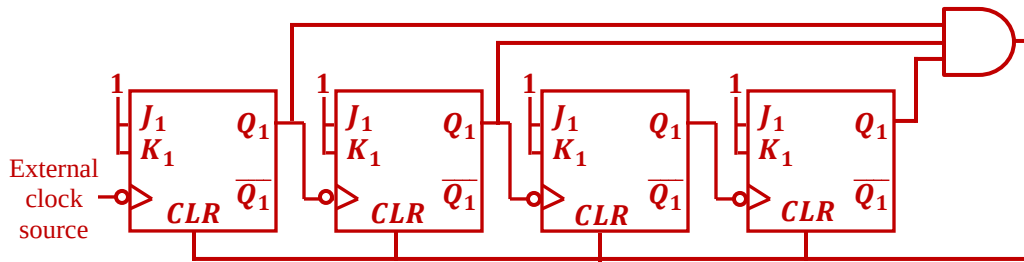
- | | | |
|---|--|---|
| 2 | What will be the sequence of outputs (Q3 Q2 Q1 Q0) generated by the counter circuit shown in the figure? Assume the initial state is (0000) ₂ . Complete the table with the resulting values. | 4 |
|---|--|---|

Ignoring the positive edge.



Q ₃	Q ₂	Q ₁	Q ₀
0	0	0	0
0	1	0	0
1	0	0	0
1	1	0	0
0	0	0	1
0	1	0	1
1	0	0	1
1	1	0	1
0	0	1	0
0	1	1	0
1	0	1	0
1	1	1	0
0	0	1	1
0	1	1	1
1	0	1	1
1	1	1	1

- 3 Design a MOD-11 counter. Draw the circuit diagram using a block diagram of negative edge triggered flip-flops with active high CLR input, and illustrate the state transitions using a timing diagram. Clearly show all states and the transition back to 0 after the maximum state is reached. 8



- 4 This is a bonus question. The total marks of the quiz will be scaled down to 8. Any surplus marks earned beyond 8 will be carried over to other quizzes where needed. 4

A committee of logic designers could not agree on whether to use NAND gates or NOR gates to build a set-reset latch. The compromise design is given in the figure.

- Fill the given characteristic table for this latch.
- Explain what is wrong with this latch?

X	Y	Q
0	0	1
0	1	1
1	0	No change
1	1	1

This latch cannot be reset.

